FINITE MODULES OVER A LOCAL RING HAVE PROJECTIVE COVERS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Nakayama's lemma. Invoked twice inside the counted proof (source lines 666-668 and 669-671) as a cross-chapter pointer. Makes the chosen map surjective and shows that proper submodules do not map onto the target.

Source reference. Original reference algebra-lemma-NAK: sources/stacks-dualizing/pinned-original-source.tex, source lines 668–668. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-NAK, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(R, \mathfrak{m}, \kappa)$ be a local ring. Any finite R-module has a projective cover.*

Proof. Let M be a finite R-module. Let $r = \dim_\kappa(M/\mathfrak{m}M)$. Choose $x_1, \ldots, x_r \in M$ mapping to a basis of $M/\mathfrak{m}M$. Consider the map $f : R^{\oplus r} \to M$. By Nakayama's lemma this is a surjection (Algebra, Lemma algebra-lemma-NAK). If $N \subset R^{\oplus r}$ is a proper submodule, then $N/\mathfrak{m}N \to \kappa^{\oplus r}$ is not surjective (by Nakayama's lemma again) hence $N/\mathfrak{m}N \to M/\mathfrak{m}M$ is not surjective. Thus f is an essential surjection. □

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